

13. Recovery Rates Clinical data concerning the effectiveness of two drugs in treating a particular condition (as measured by recovery in 7 days or less) were collected from 10 hospitals. You want to know whether the data present sufficient evidence to indicate a higher recovery rate for one of the two drugs.

- Test using the sign test. Choose your rejection region so that α is near .05.
- Why might it be inappropriate to use the Student's t -test in analyzing the data?

Hospital	Drug A				B.
	Number in Group	Number Recovered (7 days or less)	Percentage Recovered	Percentage Recovered	
1	84	63	75.0	85.4	-
2	63	44	69.8	83.1	-
3	56	48	85.7	80.2	+
4	77	57	74.0	74.5	-
5	29	20	69.0	70.0	-
6	48	40	83.3	81.5	+
7	61	42	68.9	75.4	-
8	45	35	77.8	79.2	-
9	79	57	72.2	85.4	-
10	62	48	77.4	80.4	-

Hospital	Drug B		
	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	96	82	85.4
2	83	69	83.1
3	91	73	80.2
4	47	35	74.5
5	60	42	70.0
6	27	22	81.5
7	69	52	75.4
8	72	57	79.2
9	89	76	85.4
10	46	37	80.4

$$H_0: P(A > B) = p = 0.5$$

$$H_a: P(A > B) = p \neq 0.5$$

$$n = 10, \ominus = 8, \oplus = 2.$$

$$P(X \geq 8) = P(X=8) + P(X=9) + P(X=10) = 0.055$$

$$p = 0.055 > 0.5$$

$$P(X \geq 9) = 0.0107$$

$$8 \leq k \leq 9 \Rightarrow \text{choose } 9.$$

$$RR: \{S \geq 9\}$$

6) Non-Normal Data.

Data 為比率, 被限制在 0-100%, 有可能不符 Normal Distribution

各醫院權重、變異度不均 (人數不同)